DOCUMENT 4

User Stories

Format: *As a [persona], I want [capability] so that [benefit].* Each story has testable acceptance criteria in Given / When / Then form and meets INVEST (Independent, Negotiable, Valuable, Estimable, Small, Testable). Stories flagged MUST are in the live demo.

Epic E1 · Find the right hotel

US-01 · Search hotels by my needs

MUST

As Priya, I want to search hotels by district, budget, guests, dates and the facilities I care about so that I only see hotels that fit my trip.

1. **Given** I am signed in as a guest, **when** I submit the search with any field empty, **then** I see a clear message naming the missing field and no results are loaded.
2. **Given** I choose Madurai, ₹2,500, 2 guests and valid dates, **when** I tap Find hotels, **then** the form collapses into a one-line summary and only Madurai hotels are listed.
3. **Given** check-out is the same day as check-in, **when** I search, **then** I see "Check-out must be at least one day after check-in".

Epic E1 · Feature F1 · Size S

US-02 · See the best match first

MUST

As Arun, I want each hotel to show a Best Match Score so that I can pick the best value without reading every listing.

1. **Given** search results, **when** the list loads, **then** every hotel shows a score out of 100 and the top verified hotel is labelled "Best match for you".
2. **Given** a budget of ₹2,000, **when** a hotel's room costs more, **then** it is marked "Over budget" and ranked lower than in-budget hotels with similar scores.
3. **Given** 2 nights at ₹1,500 for 2 guests, **when** I view the card, **then** the total shows ₹3,000.
4. **Given** more than 10 results, **when** I tap Next →, **then** the next 10 hotels show with "Page 2 of N".

Epic E1 · Feature F1 · Size S

US-03 · Compare hotels side by side

SHOULD

As Priya, I want to compare two or three hotels side by side so that I can decide between hotels with the same rating.

1. **Given** I tap Compare on two hotels, **when** the comparison table appears, **then** it shows score, price, total, rating, distance, matched facilities and restaurants nearby.
2. **Given** two hotels with equal ratings, **when** compared, **then** one is still highlighted "Best for you" using distance, price and facilities.

Epic E1 · Feature F4 · Size S

US-04 · Open a hotel's full details

SHOULD

As Priya, I want to open a hotel and see its rooms, facility details, parking and Wi-Fi so that I know exactly what I am getting.

1. **Given** a hotel in the list, **when** I click it, **then** a details page opens with Pre-book, Route to hotel and ← Back to hotels.
2. **Given** the hotel has a gym, **when** I view Facilities, **then** I see open hours, trainer availability and equipment; if not entered, "The hotel has not added details yet".
3. **Given** the hotel lists parking, **when** I view it, **then** I see watchman availability and car and bike slots.

Epic E1 · Feature F5 · Size S

Epic E2 · Pre-book and arrive

US-05 · Pre-book and get a Booking ID

MUST

As Priya, I want to send a pre-booking request and get a Booking ID so that my room is held without paying online.

1. **Given** a verified hotel with rooms, **when** I confirm Pre-book, **then** a unique ID in the form SZ-1234 is shown with status Pending and the estimated total.
2. **Given** a hotel whose owner is not verified, **when** I view it, **then** the Pre-book button is disabled with the reason shown.
3. **Given** any booking, **when** it is created, **then** no payment is requested anywhere.

Epic E2 · Feature F2 · Size S

US-06 · Accept or reject a request

MUST

As Ravi, I want to accept or reject pre-booking requests so that I control which rooms are held.

1. **Given** a pending request, **when** I tap Accept, **then** the status becomes Confirmed, rooms available drops by the rooms booked, and the guest sees a "confirmed" notification.
2. **Given** a pending request, **when** I tap Reject, **then** the guest sees Rejected with a suggestion to try another hotel.

Epic E2 · Feature F2 · Size S

US-07 · Verify the guest on arrival

SHOULD

As Ravi, I want to type the guest's Booking ID at the desk so that I can confirm the booking and check them in.

1. **Given** a confirmed booking for my hotel, **when** I enter its ID (any letter case), **then** I see guest name, room, nights and a Mark checked in button.
2. **Given** an ID that does not exist for my hotel, **when** I check it, **then** I see "No booking found".

Epic E2 · Feature F2 · Size S

Epic E3 · See live facility crowds

US-08 · Update a facility head count

MUST

As Ravi, I want to type how many people are in each facility so that guests see accurate crowd levels without any hardware.

1. **Given** the Gym has capacity 20, **when** I save 12, **then** it shows 60% Medium and the last-updated time becomes "just now".
2. **Given** any facility, **when** I enter a negative number or more than capacity, **then** the value is rejected with a clear message and nothing is saved.
3. **Given** I saved a count, **when** I refresh the page, **then** the saved value is still shown.

Epic E3 · Feature F3 · Size S

US-09 · See the crowd heat map for my booking

MUST

As Priya, I want to enter my Booking ID and see a heat map of the hotel's facilities so that I know which ones are quiet right now.

1. **Given** I have an active booking, **when** I enter its Booking ID on Crowd map, **then** I see the floor plan coloured green (<50%), yellow (50–79%) or red (≥80%) and a "Crowd right now" bar chart.
2. **Given** an ID that is not mine or not active, **when** I enter it, **then** I see "No active booking found with that ID" and no map.
3. **Given** staff update a count, **when** 60 seconds pass or I tap Refresh, **then** the new level is shown.

Epic E3 · Feature F3 · Size M

US-10 · Get a crowded alert and wait range

MUST

As Priya, I want a clear alert and a wait-time range when a facility is crowded so that I can decide whether to go now or later.

1. **Given** the Gym is at 85%, **when** I tap it, **then** I see "This facility is crowded" and a wait of 15–20 min (80–84% → 10–15, 85–94% → 15–20, ≥95% → 20–30).
2. **Given** a facility below 80%, **when** I tap it, **then** I see "Not crowded" and directions from Reception, with no wait time.
3. **Given** a count older than 15 minutes, **when** it is shown, **then** it is hatched and marked "This data may be outdated".

Epic E3 · Feature F3 · Size S

Epic E4 · Wait smarter

US-11 · Take a free virtual queue token

SHOULD

As Priya, after check-in, I want a free virtual token for a crowded facility so that I can do something else instead of standing in line.

1. **Given** I am checked in and the Gym is High, **when** I tap Get free virtual token, **then** I get a code like Q-017 and my queue position.
2. **Given** my booking is only pending or confirmed, **when** I view a crowded facility, **then** I see the wait range but no token button.
3. **Given** staff tap Call next until I am next, **when** that happens, **then** I see "Your turn is coming up".

Epic E4 · Feature F6 · Size S

US-12 · Get a suggestion for the wait

SHOULD

As Priya, I want StayZen to suggest a quiet facility that fits my wait so that I use the time well and do not miss my turn.

1. **Given** a 15–20 min wait for the Gym, **when** I wait, **then** I see up to two quiet facilities whose usual duration is 20 minutes or less.
2. **Given** a suggestion, **when** I tap Accept, **then** the route to that facility is shown; **when** I tap Dismiss, **then** it disappears.

Epic E4 · Feature F6 · Size S

Epic E5 · Trusted listings

US-13 · Register my hotel

SHOULD

As Ravi, I want to create a manager account together with my hotel so that my hotel can appear on StayZen.

1. **Given** the Hotel Management sign-in, **when** I complete "Register your hotel", **then** I am signed in and my hotel is saved as Pending.
2. **Given** my hotel is Pending, **when** a guest searches, **then** my hotel does not appear.
3. **Given** an email that already exists, **when** I register, **then** I see "An account with this email already exists".

Epic E5 · Feature F7 · Size S

US-14 · Verify hotels and spot outdated data

SHOULD

As the Stays Admin, I want to verify new hotels and see outdated crowd feeds so that guests only see trustworthy information.

1. **Given** a pending hotel, **when** I tap Verify & approve, **then** it gets the Verified badge, appears in search and accepts pre-bookings.
2. **Given** a hotel changes a room price, **when** I open Submitted updates, **then** I see the change and can mark it reviewed.
3. **Given** a verified hotel's facility not updated for over 15 minutes, **when** I open Data freshness, **then** it is listed with its age.

Epic E5 · Feature F7 · Size S

US-15 · Finish my trip and give feedback

COULD

As Priya, I want to finish my trip and rate my stay so that the hotel and StayZen can improve.

1. **Given** a checked-in booking, **when** I tap Finish trip, **then** the status becomes Completed and the 8-item feedback form opens.
2. **Given** the form, **when** I submit without rating every item, **then** I see "Please rate every item"; feedback can be sent only once per booking.

Epic E5 · Feature F10 · Size S

Epic E6 · Help in my language

US-16 · Ask the assistant in Tamil or English

COULD

As Arun, I want to ask the StayZen assistant questions in Tamil by text or voice so that I understand answers in my own language.

1. **Given** I switch to தமிழ், **when** I ask "ஜிம்மில் கூட்டம் உள்ளதா?", **then** the reply is in Tamil with the Gym's current level.
2. **Given** I ask something StayZen has no data for, **when** it replies, **then** it says it needs confirmation and never invents prices or offers.
3. **Given** Voice is on, **when** a reply arrives, **then** it is read aloud.

Epic E6 · Feature F8 · Size S

US-17 · Reach emergency help quickly

COULD

As Priya, I want emergency numbers and the nearest hospital or police station in one tap so that I can get help fast in an unfamiliar town.

1. **Given** the Safety tab, **when** it opens, **then** 112, 100, 108 and 101 are shown at the top.
2. **Given** a booking, **when** I tap Find nearest & navigate for Hospitals, **then** Google Maps opens searching near my hotel's location.

Epic E6 · Feature F9 · Size S

DOCUMENT 5

Traceability matrix

Every story traces to an epic, a PRD feature and an MRD need. Demo items are the Must-Have stories.

STORY	EPIC	PRD FEATURE	MRD NEED	PRIORITY	IN DEMO
US-01 Search hotels by my needs	E1	F1	N1	MUST	Yes
US-02 See the best match first	E1	F1	N1	MUST	Yes
US-03 Compare hotels	E1	F4	N1	SHOULD	Yes
US-04 Hotel full details	E1	F5	N1, N4	SHOULD	Yes
US-05 Pre-book and get a Booking ID	E2	F2	N2	MUST	Yes
US-06 Accept or reject a request	E2	F2	N2	MUST	Yes
US-07 Verify the guest on arrival	E2	F2	N2, N4	SHOULD	Yes
US-08 Update a facility head count	E3	F3	N3	MUST	Yes
US-09 Crowd heat map for my booking	E3	F3	N3	MUST	Yes
US-10 Crowded alert and wait range	E3	F3	N3	MUST	Yes
US-11 Free virtual queue token	E4	F6	N3	SHOULD	Yes
US-12 Suggestion for the wait	E4	F6	N3	SHOULD	If time
US-13 Register my hotel	E5	F7	N4	SHOULD	If time
US-14 Verify hotels, spot outdated data	E5	F7	N4	SHOULD	Yes
US-15 Finish trip and feedback	E5	F10	N4	COULD	If time
US-16 Assistant in Tamil or English	E6	F8	N5	COULD	Yes
US-17 Emergency help	E6	F9	N5	COULD	If time

Data honesty statement

- **Real:** hotel and restaurant names, locations, phone numbers, websites and star tags from OpenStreetMap (Overpass API), collected by the team.
- **Sample values in the prototype:** room prices, room counts, facility details, parking and Wi-Fi details for OpenStreetMap hotels, and starting crowd counts. These are replaced when a hotel manager registers and edits their hotel.
- Crowd levels come from head counts typed by staff. No cameras or sensors are used, and no trained prediction model is claimed.